"Good [morning/afternoon]. Today I'd like to walk you through the **AQI Intelligence Portal** — a satellite-based air quality monitoring system built for Abu Dhabi Emirate, covering January 2022 to December 2024.

Before we look at the platform itself, I want to briefly explain what sits behind it — because the real power of this portal is not just in the data, but in making that data understandable.

The system processes air quality through a **four-stage pipeline:**

First, **data acquisition** — satellite observations are downloaded from Google Earth Engine, pulling high-resolution measurements from ESA's Sentinel-5P and NASA's MODIS, alongside global atmospheric reanalysis from ECMWF's CAMS system.

Second, **data preparation** — raw satellite signals are converted from column densities into actual ground-level mass concentrations, reprojected onto a uniform 1-kilometer grid, and missing pollutants are derived from aerosol optical depth using regression models.

Third, **ground station normalization** — the satellite estimates are calibrated against 20 physical EAD monitoring stations using Inverse Distance Weighting interpolation. This single step alone improves accuracy by 15 to 25 percent.

And fourth, **temporal aggregation and export** — all sources are merged, resampled across time periods from hourly to yearly, AQI values are computed, and the results are exported as georeferenced raster files ready for spatial analysis.

Now — all of this processing can be done. The science is sound. The data is accurate. But raw numbers and raster files alone do not tell a story. You can process terabytes of satellite data, but without the right visualization, you cannot see *where* air quality is (failing) deteriorating, *when* seasonal patterns emerge, or *how much* one period improved compared to another.

That is precisely what this portal delivers — it transforms a complex, multi-source scientific pipeline into an interactive, visual experience that allows analysts, planners, and decision-makers to ask questions and find answers directly on the map.

Let me show you how."

Look & Feel — Overall Layout

"The application is built as a full-screen, two-section layout. The **upper section** is the interactive ESRI map, displayed on a high-resolution satellite basemap that immediately gives geographic context to the emirate. The **lower section** is the analytics panel, which updates dynamically based on the active tab.

In the top-left of the map, you'll notice the portal branding — the title **'Air Quality Intelligence Portal'** — alongside an information button. In the top-right, the SCAD logo is (fixed) anchored as an institutional identifier.

The bottom panel can be **collapsed or expanded** at any time using the toggle button, giving the map more screen real estate when needed."

Map Tools (ArcGIS Widgets)

"The map comes with a full set of ESRI map native controls, all accessible from the top-left toolbar:

- **Basemap Gallery** — allows switching between satellite, dark gray, light gray, and streets basemaps to suit the presentation context.
- **Layer List** — provides visibility control for all layers. Each layer also includes an **opacity slider**, so the AQI heatmap transparency can be adjusted against the satellite imagery to fine-tune readability.
- **Legend** — a live legend panel that reflects the currently displayed pollutant and its color-coded AQI classification.
- **Zoom controls** — standard zoom in/out for map navigation.

On the **swipe view**, the right-side map also has its own independent layer list with the same opacity control."

Identify / Feature Info Tool

"One of the most powerful interaction features is the **click-to-identify tool**. In any view, clicking anywhere on the map triggers two things simultaneously:

1. The **1-kilometer grid cell** at that location is highlighted with a cyan outline, giving precise spatial context to the query.
2. A **Feature Info panel** slides in from the right side, displaying the pollutant values for that exact grid cell — AQI, SO_2, NO_2, CO, Ozone, PM2.5, and PM10 — with their units.

If no data exists at the clicked location, a toast notification informs the user immediately. The info panel can be dismissed with the close button, which also clears the grid highlight."

Tab 1 — Monthly View

"The default tab is the **Monthly View**. Here, the analyst selects a year from a dropdown and navigates months using a **stepped slider**. Clicking play triggers an **animated playback** — the map automatically steps through each month in sequence, revealing how air quality evolves across the emirate over the year.

The analytics panel shows three sections:

- A **pie chart** showing the distribution of AQI categories — Good, Moderate, Unhealthy, and so on — for the selected period.
- A **summary statistics card** for the overall AQI, showing minimum, maximum, and mean values.
- **Individual pollutant cards** for all six pollutants — SO_2, NO_2, CO, Ozone, PM2.5, and PM10 — each showing their min and max values."

Tab 2 — Yearly View

"The **Yearly View** shifts the temporal resolution to annual aggregates. Instead of a slider, year selection is done through **year buttons** — 2022, 2023, 2024 — making it fast to jump between years for a high-level comparison.

The same pie chart and statistics panel update to reflect the full-year aggregated data, giving a quick performance overview at an annual level."

Tab 3 — Monthly Analysis (Side-by-Side Comparison)

"The **Monthly Analysis** tab is the split-screen comparison mode. The map divides into two synchronized panels — **Baseline on the left, Current on the right** — both displaying the AQI heatmap for independently selected months and years.

Both maps are **fully synchronized**: panning or zooming on one automatically mirrors the other, so spatial alignment is always maintained. Each side has its own year dropdown, month slider, and independent play button.

The central statistics panel shows the **percentage change** between the two periods for AQI and all six pollutants. A red upward arrow indicates deterioration; a green downward arrow indicates improvement — making trend analysis immediately visual and intuitive."

Tab 4 — Ground vs Satellite Data

"This tab validates the satellite model against real ground measurements. On the left, the analyst selects a year and month, then chooses one of the **20 EAD monitoring stations** from the scrollable list. The map zooms to the selected station, which is highlighted with a pulsing cyan marker.

The chart panel on the right shows a **grouped bar chart** comparing EAD ground readings against satellite-derived values for every pollutant at that station and period.

Above the chart, two color-coded AQI summary cards display the ground AQI and the satellite AQI side by side — color-classified from Good to Hazardous — so the calibration accuracy is immediately visible. A disclaimer tooltip is also available for methodological transparency."

Tab 5 — District Analysis

"The **District Analysis** tab shifts the focus from grid-level to administrative-level analysis. The analyst selects a year, a month, and then chooses a district from the scrollable list — or simply **clicks directly on the map** to select a district, which triggers an automatic zoom to that district's boundary.

The map renders each district with an **AQI-colored renderer**, so the spatial distribution of air quality categories across Abu Dhabi, Al Ain, and Al Dhafra is immediately apparent.

The right panel shows the district's AQI classification card and a **bar chart of all six pollutant concentrations** for the selected period, enabling district-level reporting and comparison."

Info Panel

"Finally, the **information button** in the top-left opens a full-screen informational modal. This covers the AQI methodology, the data sources — Sentinel-5P, MODIS, and CAMS — the processing pipeline, validation accuracy results of 93 to 97 percent, and guidance on how to use the portal. It serves as both an onboarding guide and a technical reference."

Closing

"In summary, this portal brings together satellite data fusion, ground-station calibration, and an intuitive geospatial interface into a single tool — enabling environmental agencies, policymakers, and researchers to monitor, compare, and analyze air quality across Abu Dhabi with confidence and precision. Thank you."